Arduino UNO R4 WiFi TITO Deluxe

Marc R. Davis

Project: https://github.com/marcrdavis/ArduinoTITO-PlayerTracking

June 23, 2026 – Version 4.0.20260623

HARDWARE REQUIRED

Arduino UNO R4 WiFi
https://www.amazon.com/Arduino-UNO-WiFi-ABX00087-Bluetooth/dp/B0C8V88Z9D

MAX3232 Serial port (or compatible)
https://www.amazon.com/gp/product/B083L99CGZ

SD Card Shield for Arduino (or a Micro SD TF Card Module; SPI) RECOMMENDED
https://www.amazon.com/HiLetgo-Stackable-Card-Shield-Arduino/dp/B006LRR0IQ

1' USB Cable
https://www.amazon.com/GHNTJAP-SuperSpeed-Charging-Android-Controller/dp/B0D541N33M/

Four 2" DuPont M-F jumper wires (6 more if you are using the standalone SD module)

Serial cable to connect to game

- For IGT Games: 5-pin Dupont to Male DB9 Serial pigtail (I reused the one from the BETTORSlots TITO board)
- https://www.amazon.com/gp/product/B081GJR1MN

5 Watt or greater USB Power Brick

Compatible slot machine

- Tested on IGT S2000/GameKings/AVP, Bally, WMS and Konami machines; will probably work on others based on the SAS 6.x protocol; you will need a compatible cable to connect to the serial port on the machine

SOFTWARE REQUIRED

Latest version of Arduino IDE
https://www.arduino.cc/en/software

BEFORE YOU BEGIN

These instructions assume familiarity with electronics, coding and slot machine configurations. If you are new to any of these then this project may not be for you. My instructions were not written for beginners. You may damage your game or the Arduino/TITO hardware if you do not understand what you are doing.

WIRING

The Serial board needs to be wired as follows:

Serial	Arduino
VCC	5V
RXD	Data Pin 0
TXD	Data Pin 1
GND	GND

If you are using the optional SD Card Module (not the shield) it needs to wired as follows:

SD	Arduino
VCC	5V
GND	GND
CLK/SCK	Data Pin 13
DO/MISO	Data Pin 12
DI/MOSI	Data Pin 11
CS/SS	Data Pin 10

You will need to power the Arduino board from the Accessory Outlet in the base of the slot machine using a USB power brick (5W or greater).

- Do not power the board from any built-in USB port on the game – it will not provide enough power

INITIAL SETUP

- Assumes you have the Arduino IDE setup, the board and COM port settings are correct
- Install version 1.2.2 of the Arduino UNO R4 Boards from within Boards Manager
    - You must update two files to enable 9-bit Serial support. The updated files are located in the 'UNO R4 WiFi 9-Bit Serial' folder within the package
    - Replace HardwareSerial.h in C:\Users\<YOURID>\AppData\Local\Arduino15\packages\arduino\hardware\renesas_uno\1.2.2\cores\arduino\api
    - Replace Serial.cpp in C:\Users\<YOURID>\AppData\Local\Arduino15\packages\arduino\hardware\renesas_uno\1.2.2\cores\arduino
- The following libraries and versions are required:
    - WiFiS3 (latest)
    - SD (latest)
    - SPI (latest)
    - ArduinoGraphics (latest)
    - Arduino_LED_Matrix (latest)

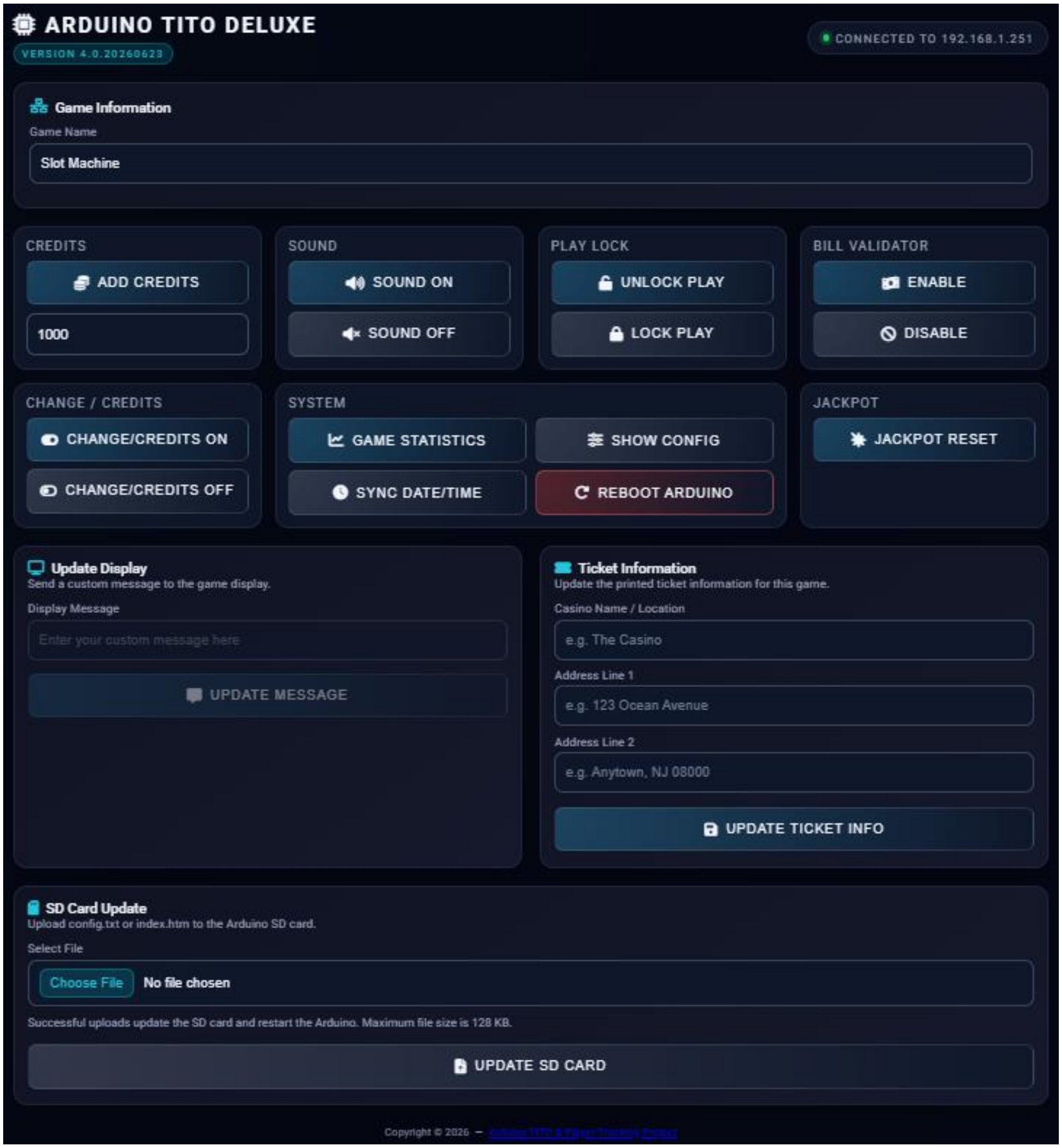
    *Can be downloaded via the Arduino IDE Library Manager

- Connect the board to your computer via USB and load the ArduinoTITODeluxeR4WiFi sketch
- **The SD card is optional, but highly recommended; when present the board will attempt to read its configuration from the SD card. If the config.txt file is not present or if settings are missing the hard-coded defaults will be used.**
    - If you want to use the web UI then the SD card is required. Alternatively, you could control the machine from the Game Manager Windows app.
- [SD Card] Format the MicroSD Card as FAT32
- [SD Card] Edit the included config.txt file with your settings and preferences; then copy it and the index.htm file to the SD card; see the Arduino TITO and Player Tracking Project - File Formats.txt for details on the supported parameters
- **If you are not using the SD card you must modify the following settings in the sketch before you download it to the Arduino UNO R4 WIFI:**
    - ChangeToCredits – set to 1 to add credits via the Service/Change button
    - changeCredits – set to the number of credits to add on each push of the Service/Change button
    - noSDCard – Set to 1 to disable the SD card, if present
    - useDHCP – set to 1 to get a unique IP address from the network; you will need to discover the address by checking your router after the device is connected
    - ssid – set to the name of your WiFi
    - pass – set to the password for your WiFi
    - ip – set a unique IP address for your network; required if useDHCP is set to 0
- Download the sketch to the Arduino; wait until complete
- Unplug the Arduino from the PC
- Mount the hardware into the slot machine's lower cabinet

- Connect the board to the game’s serial port using the DB9-to-DuPont cable or existing compatible cable
- Connect the USB power to the machine’s accessory outlet
- Power on the machine and test
    - Assumes you are replacing an existing TITO board or have already setup your machine per the instructions later in this document

REMOTE ACCESS

The web interface allows you to control various aspects of your game by using a web browser.



Most functions are self-explanatory; The Change/Credits feature allows you to enable adding credits by pressing the Change (or Service) button. The number of credits added on each press is set in the sketch (or the config file) prior to loading it onto the Arduino.

The Update Ticket Info option allows you to change the information printed on the cash-out ticket.

Some features are not available on all games or versions of the Arduino TITO board.

CONFIGURING YOUR IGT MACHINE (S2000/GameKing)

Before getting started with setting up your IGT machine for TITO please ensure your Bill Validator is working correctly and your Ticket Printer can print clear and legible tickets. You will also need a Keychip appropriate for your type of machine. These instructions assume familiarity with the Keychip process. Note – keychips and menu option locations vary from model-to-model. Please consult your IGT user manual or me if you have questions.

1. Clear any credits off your machine
2. Keychip your machine
3. Once in the Keychip Menu, ensure your Denomination, Devices, Limits and Game settings are as you want them
4. Setup the Comm Options as follows
    a. IGT SAS Primary Channel = Channel 3
    b. SAS Secondary = Off
    c. Bally Miser = Off
    d. Progressive Link = 7
    e. WAMM 1.0 = Off
5. Setup the Validation/Redemption Options as follows
    a. Validation = System Validation
    b. Redemption = SAS Redemption
6. Setup the IGT SAS Options as follows
    a. System Bonusing = SAS Legacy
7. Setup the SAS Channel (Primary Channel) Options as follows
    a. Address = 1
    b. Legacy Bonus = Enabled (X)
    c. Validation = Enabled (X)
8. Setup the Machine Terminal Options as follows
    a. Voucher Limit Follows = Credit Limit
9. Save all options and make any other changes you wish before pressing Return to Game to exit the keychip menu

Test the game by inserting money and then pressing Cash Out to generate a ticket. The ticket serial number should match the number of credits you inserted. Insert the ticket into the machine, it should accept it for the same number of credits.

CONFIGURING YOUR IGT MACHINE (AVP)

Please see this video for how to configure your AVP Game:
https://www.youtube.com/watch?v=JKjyeFQPltA

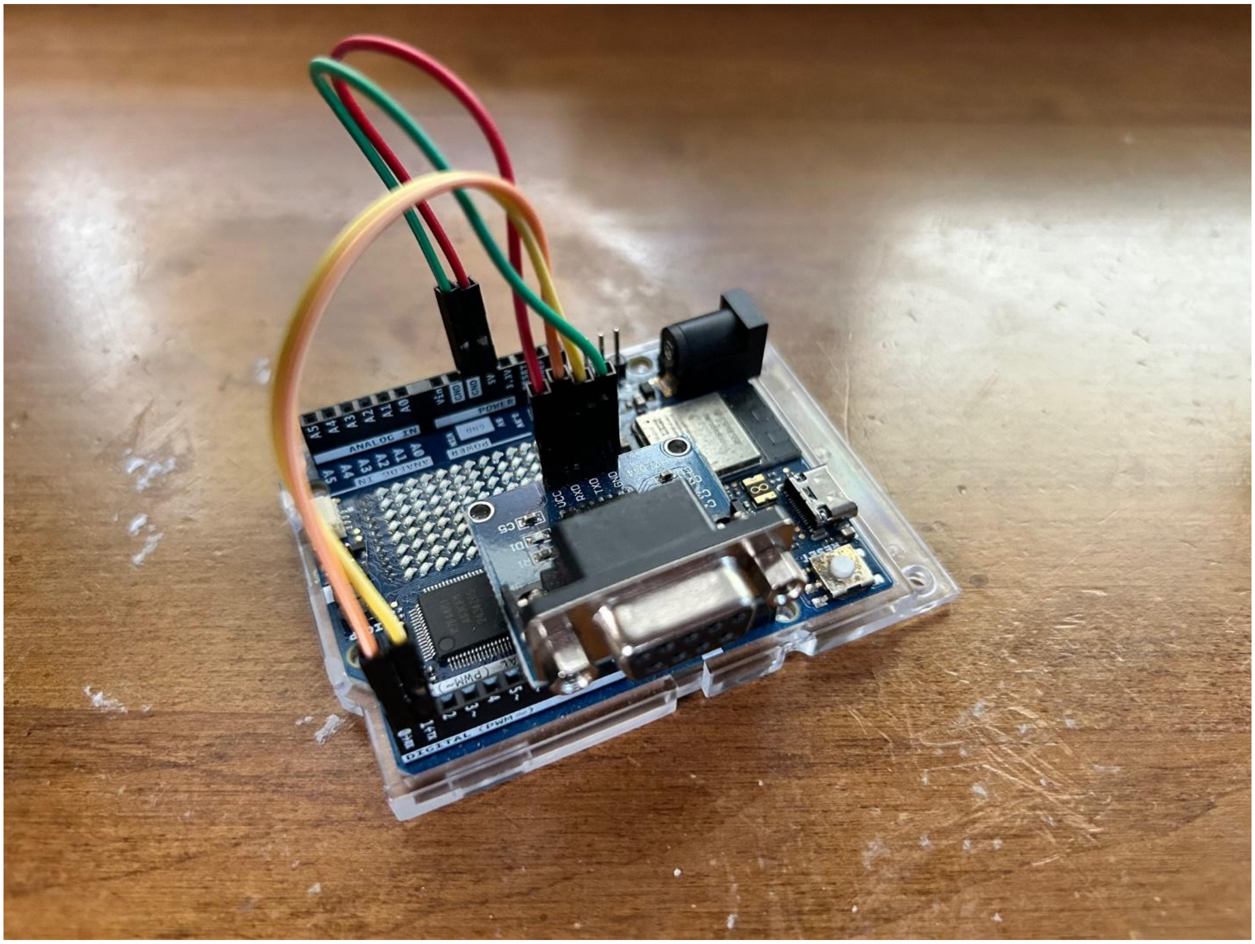
NOTES AND COMMENTS

- The board will output diagnostic info via the USB-C serial port when powered by a laptop running the Arduino IDE Serial Monitor
- The Matrix display on the R4 board will display OK when the board is running normally and ERR when a network or serial error has occurred. Use the Serial Monitor (above) to diagnose the issue

DISCLAIMER

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

Figure 1 – Mockup



Example of R4 board wiring

FIGURE 2: IGT Serial Cable

IGT S2000/GameKing Serial Pinout

DB9 Male	Signal direction	S2000 J82
3 Receive Data	←	1 Transmit Data
2 Transmit Data	→	2 Receive Data
5 Signal Ground	---	5 Signal Ground

Option: DB9 to Dupont Serial Cable: https://www.amazon.com/gp/product/B081GJR1MN